

## AIRSTREET · PROBLEM STATEMENT

# AirStreet problem statement

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## DOCUMENT CONTROL

|             |                                            |         |                |
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| Project TRL | <b>TRL 2</b> Technology concept formulated |         |                |
| Repository  | github.com/BoujeeEnjinia1701/airstreet     |         |                |
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## REVISION HISTORY

| Version | Date       | Author       | Change                                                               |
|---------|------------|--------------|----------------------------------------------------------------------|
| 0.2     | 2026-09-25 | Amish Chadha | Populate to TRL 2 (problem, users, context, constraints, prior work) |
| 0.1     | 2026-09-25 | Amish Chadha | Initial scaffold                                                     |

# AirStreet problem statement

Regulatory air monitors are few and far apart, missing the streets where people breathe traffic pollution.

## ■ The problem

Traffic pollution varies sharply over short distances, but official monitoring is designed to describe whole cities. When Google Street View cars carrying reference-grade instruments drove every street in a 30 km<sup>2</sup> area of Oakland, California, nitrogen dioxide (NO<sub>2</sub>), nitric oxide and black carbon varied by up to 5 to 8 times within single city blocks (Apte et al., 2017). A single fixed station cannot show that pattern, and many cities have no station at all. Only 12 % of India's 4,041 census cities and towns have any air quality monitoring (CSE, 2023), and in 2020 OpenAQ found no public national government monitoring program in 13 populous countries, including Pakistan, Nigeria, Ethiopia, Kenya and Uganda (OpenAQ, 2020).

The health stakes are large. Ambient air pollution caused an estimated 4.2 million premature deaths worldwide in 2019 (WHO). NO<sub>2</sub> comes mainly from burning fuel in cars, trucks, buses, power plants and off-road equipment (US EPA), so it is highest on busy streets, and more than 45 million people in the United States live within about 90 m (300 ft) of a major transportation facility (US EPA).

Low-cost sensors could fill the gap between stations, but raw readings are not trusted. Uncorrected PurpleAir particle sensors overestimated PM<sub>2.5</sub> by about 40 % across the United States until a humidity-aware correction cut the error from 8 to 3 µg/m<sup>3</sup> (Barkjohn et al., 2021). Electrochemical NO<sub>2</sub> sensors respond to temperature, humidity and other gases, and laboratory calibration alone does not correct this; in a Pittsburgh study only field-calibrated models reached a mean absolute error of about 3.5 ppb for NO<sub>2</sub> (Zimmerman et al., 2018).

AirStreet addresses one specific gap: an open, pole-mounted node that measures PM<sub>2.5</sub> and NO<sub>2</sub> at street level, with a calibration method stated up front, so a city, school or community group can build a neighborhood map that others can check.

## ■ Users and context

| User                                               | Need                                                                                                 | Context                                                                |
|----------------------------------------------------|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Residents and community groups                     | See pollution on their own streets, near schools and clinics, and track whether it changes           | Street poles on residential and arterial roads                         |
| Municipal environment and transport teams          | Evaluate interventions such as low-traffic streets, school streets, bus corridors and freight routes | 10 to 100 nodes per district, alongside at least one reference station |
| Schools and universities                           | Teaching and research on exposure, with raw data and a documented method                             | Campus and nearby streets                                              |
| Street furniture owners (utilities, city lighting) | A small, light device that does not interfere with the pole or its wiring                            | Street light and signal poles                                          |
| Researchers and data platforms                     | Raw signals and calibration versions, so data can be reprocessed                                     | Open data portals                                                      |

## ■ Operating environment

- Mounted on a street light or signal pole, inlets about 3 m above the sidewalk (proposed), on the street side and within a few meters of the kerb.
- Outdoor, in sun, rain, dust and traffic spray; temperate to hot humid climates (about -10 to 50 °C, 10 to 100 % RH, estimate to be confirmed with partners).
- No mains power: runs on the FieldNode solar and battery core.
- Data over LoRaWAN to a community or municipal gateway (for example the lab's TwinKit gateway) or a public network.

## ■ Constraints

- Garage-buildable prototype, about \$180 USD (the current estimate exceeds this; see [03-requirements.md](#)).
- Built on FieldNode, the lab's shared outdoor core, with its power budget and sensor port pinout.
- Temperature, humidity and particle checks on CalRig. CalRig does not cover NO<sub>2</sub>, so NO<sub>2</sub> must be calibrated by field collocation with a reference station.
- Levels only: no images, audio or personal data collected or sent.
- Mounting only with the pole owner's permission, without drilling the pole or opening its electrical access.

## ■ Out of scope

- Regulatory compliance monitoring. AirStreet data are indicative and do not replace reference stations.
- Other pollutants (ozone, carbon monoxide, black carbon, ultrafine particles) in this version.
- Personal exposure devices and indoor use.
- Health advice or medical claims.

## ■ Prior work

- Street-scale mapping with reference instruments on vehicles showed block-level variation ([Apte et al., 2017](#)).
- Community and city networks of low-cost sensors already operate, for example AirQo, which reports more than 400 PM<sub>2.5</sub> sensors across 14 African countries ([AirQo](#)), and Breathe London, a sensing network developed with the Mayor of London ([Breathe London](#)).
- The US EPA published performance testing protocols for NO<sub>2</sub>, CO and SO<sub>2</sub> sensors in February 2024, including field evaluation alongside regulatory monitors ([US EPA](#)).
- Humidity correction for optical particle sensors ([Barkjohn et al., 2021](#)) and field calibration of electrochemical gas sensors ([Zimmerman et al., 2018](#)) are established methods that AirStreet adopts rather than reinvents.

What is missing is an open, documented node design that pairs these sensors with a stated, repeatable calibration route and publishes raw signals.

## ■ Open questions

- Which city and partner first, and which reference station can host a collocation? Proposed, awaiting Amish.
- Inlet height of 3.0 m or lower, closer to the breathing zone? Proposed 3.0 m, awaiting Amish.
- Is NO<sub>2</sub> accuracy of about 5 ppb hourly useful enough for the first partner's questions? To be tested with users.

## ■ User research and co-design

This design is for communities the author is not part of, so requirements come from the people who will use it.

- [ ] Identify a local partner organization (Helpful Engineering network, NGO or university)
- [ ] Run co-design sessions with intended users; record who, where and what was learned
- [ ] Validate load, distance, terrain and cost assumptions in the field
- [ ] Revise requirements (REQ) from findings before freezing the design